Sustainability for Robotics Checklist

This 10-point checklist is derived from a draft of the BS8622 Guide to Robot Sustainability.

Not all points on this checklist may be relevant to the scope of the design. Where it is not relevant, this should be justified.

1. **Materials**
   Identify the types of materials currently used in the design and possible ways to incorporate more environmentally friendly materials into the robot.

2. **Software**
   How can software overheads (power, communications, data storage, training) be minimized or made more sustainable?

3. **Energy**
   Has energy efficiency been considered within the design? What could be done to reduce energy consumption? What is the source of energy for the robot and are there more environmentally friendly options?

4. **Waste**
   Identify if any waste products are generated by the robot during its operation and consider measures to reduce and handle them to minimize the environmental impact.

5. **Emissions**
   Identify possible sources of pollution generated by the robot and consider measures to reduce and handle them to minimize the environmental impact.

6. **Communications**
   Consider how much data is being transmitted to and from the robot. Is it all necessary? What are the data storage implications of data transmitted from the robot?

7. **Modularity**
   How modular is the robot design? Can items be easily replaced or repaired? Consider both the hardware and software.

8. **Location/Placement**
   Are there any sustainability considerations with the deployment of the robot? Does the robot need to be transported using vehicles? If so, what is the most sustainable method of transportation?

9. **Maintenance**
   How is the robot maintained? What checks are taken to ensure it doesn't break mid-mission? Does the robot have diagnostic software which monitors its performance, which can be used to identify when maintenance needs to be conducted?

10. **Repurposing**
    Consider how the robot may be re-used at the end of the project. Can it be re-purposed for other activities?

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